Q| Show that $u(x,y) = \sin x \cosh y$ is a harmoic func.

$u(x,y) = \sin x \cosh y$

$\frac{\partial u}{\partial x} = \cos x \cosh y$ | $\frac{\partial u}{\partial y} = \sin x \sinh y$

$\frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y$ | $\frac{\partial^2 u}{\partial y^2} = \sin x \cosh y$

The laplace eq:

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\cancel{\sin x \cosh y} + \cancel{\sin x \cosh y} = 0$ is satisfied

Therefore $u(x,y) = \sin x \cosh y$ is a harmonic func.

Linearization:

The Linearization of a func $f(x,y)$ at a point (x_0, y_0) where $f(x,y)$ is diffrentiable

$$L(x,y,z) = f(x_0,y_0,z_0) + f_x(x_0,y_0,z_0)(x-x_0) + f_y(x_0,y_0,z_0)(y-y_0) + f_z(x_0,y_0,z_0)(z-z_0)$$

for plane

The approximation:

$f(x,y) \approx L(x,y)$ is the standard approx at (x_0,y_0)

Q| find the Linearization of $f(x,y) = x^2 - xy + \frac{1}{2}y^2 + 3$ at the point $(3,2)$

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

$L(x,y) = f(3,2) + f_x(3,2)(x-3) + f_y(3,2)(y-2)$

$f(x,y) = x^2 - xy + \frac{1}{2}y^2 + 3$

$f(3,2) = (3)^2 - (3(2)) + \frac{1}{2}(2)^2 + 3$

$= 9 - 6 + 2 + 3$

$= 8$

$f_x = 2x - y$ | $f_y = -x + y$

$f_x(3,2) = 6 - 2 = 4$ | $f_y(3,2) = -1$

$L(x,y) = 8 + 4(x-3) - 1(y-2)$

$= 8 + 4x - 12 - y + 2$

$L(x,y) = 4x - y - 2$

$L(3,2) = 4(3) - 2 - 2$

$= 12 - 4$

$\boxed{L(3,2) = 8}$

Q/- $f(x,y,z) = x^2 - xy + 3\sin z$ at $(3,1,0)$

$f(3,1,0) = (3)^2 - 3(1) + 3\sin 0$

$f(3,1,0) = 6$

$f_x = 2x - y$ | $f_y = -x$ | $f_z = 3\cos z$

$f_x(3,1,0) = 2(3) - 1 = 5$ | $f_y(3,1,0) = -3$ | $f_z(3,1,0) = 3$

$L(x,y,z) = f(3,1,0) + f_x(3,1,0)(x-3) + f_y(3,1,0)(y-1) + f_z(3,1,0)(z-0)$

$L(x,y,z) = 6 + 5(x-3) + (-3)(y-1) + 3z$

$= 6 + 5x - 15 - 3y + 3 + 3z$

$L(x,y,z) = 5x - 3y + 3z - 6$

$L(3,1,0) = 5(3) - 3(1) + 3(0) - 6$

$= 15 - 3 + 0 - 6$

$\boxed{L(3,1,0) = 6}$

The Chain Rule:

we used the chain rule when $w = f(x)$ was a differentiable func of x and $x = g(t)$ differentiable func of t and the chain rule said that $\frac{dw}{dt}$ could be calculated with formula

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt}$$

if $w = f(x,y)$ is differentiable and x, y are diff func of t then w is a differentiable func of t.

$$\boxed{\frac{dw}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}}$$

Ex: use the chain rule to find derivative $w = xy$ w.r.t along path $x = \cos t$, $y = \sin t$ what is the derivative value at $t = \frac{\pi}{2}$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dt}$$

$$= y(-\sin t) + x\cos t$$

$$\frac{dw}{dt} = -\sin^2 t + \cos^2 t$$

$$\left.\frac{dw}{dt}\right|_{t=\frac{\pi}{2}} = -1$$

Q/ Find $\left.\frac{dw}{dt}\right|_{t=0}$ of

$w = xy + z \qquad x = \cos t,\ y = \sin t,\ z = t$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial w}{\partial z}\cdot\frac{dz}{dt}$$

$$= y(-\sin t) + x(\cos t) + 1$$

$$= -\sin^2 t + \cos^2 t + 1$$

$$\left.\frac{dw}{dt}\right|_{t=0} = 0 + 1 + 1 = 2$$

Chain Rule (Most Imp w.r.t exam)

Chain rule for two indep var and three intermediate var:

Suppose $w = f(x,y,z)$, $x = g(r,s)$, $y = h(r,s)$, $z = k(r,s)$

if all four funcs are differentiable. then w has the partial derivative w.r.t r and s given by the formula.

$$\boxed{\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial r}}$$

$$\boxed{\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial s}}$$

Ex: ★ if $w = f(x,y)$, $x = g(r,s)$, $y = h(r,s)$ then.

$$\boxed{\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial r}}$$

$$\boxed{\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial s}}$$

★ if $w = f(x)$ $\quad x = g(r,s)$

Then $\boxed{\frac{\partial w}{\partial r} = \frac{dw}{dx}\cdot\frac{\partial x}{\partial r}}$ & $\boxed{\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial s}}$

Express $\frac{\partial w}{\partial r}$ & $\frac{\partial w}{\partial s}$ in terms of r & s if $w = x - 2y + z^2$.

$x = r/s$, $y = r^2 + \ln s$, $z = 2r$.

i) $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}$

$\because z = 2r$

$= 1 \times \frac{1}{s} + (-2)(2r) + 2z \cdot 2 = \frac{1}{s} - 4r + 4z = \frac{1}{s} - 4r + 4(2r)$

$= \boxed{\frac{1}{s} + 4r}$

ii) $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s}$

$= 1 \times \frac{-r}{s^2} - 2 \times \frac{1}{s}$

$= \boxed{\frac{-r}{s^2} - \frac{2}{s}}$

$x = \frac{r}{s}$ $\quad r \; s^{-1}$

$= r \cdot -1 \cdot s^{-2}$

$\frac{-r}{s^2}$

Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r & s if $w = x^2 + y^2$, $x = r - s$, $y = r + s$

i) $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r} = 2x \cdot 1 + 2y \cdot (1) = 2x + 2y$

$= 2(x+y) = 2(r - \cancel{s} + r + \cancel{s}) = 2(2r) = \boxed{4r}$

ii) $\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} = 2x \cdot (-1) + 2y(1) = -2x + 2y$

$= 2(y - x) = 2(\cancel{r} + s - \cancel{r} + s) = 2(2s) = \boxed{4s}$

Implicit: The func $f(x,y)$ is differentiable and the eq $f(x,y) = 0$ defines y implicit as func of x say $y = h(x)$ Since $w = f(x,y) = 0$

$\boxed{\frac{dw}{dx} = f_x \frac{dx}{dx} + f_y \frac{dy}{dx} = 0}$

$f_x + f_y \frac{dy}{dx} = 0$

$f_y \frac{dy}{dx} = -f_x$

$\frac{dy}{dx} = \frac{-f_x}{f_y}$ if $f_y \neq 0$

Ex: Find $\frac{dy}{dx}$ if $x^2+\sin y-2y=0$

$$\frac{dy}{dx} = \frac{-f_x}{f_y}$$

$f_x = 2x$

$f_y = \cos y - 2$

$$\boxed{\frac{dy}{dx} = -\frac{2x}{\cos y-2}}$$

Ex: $w=x^2+y^2$, $x=\cos t$, $y=\sin t$ $t=\pi$, $\frac{dw}{dt}$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dt}$$

$$= 2x\cdot(-\sin t) + 2y(\cos t)$$

$$= -2\sin t\cos t + 2\sin t\cos t = \boxed{0}$$

~~$\cos 90$~~

$\because \sin 180 = 0$

Q1- $w=\frac{x}{z}+\frac{y}{z}$, $x=\cos^2 t$, $y=\sin^2 t$, $z=\frac{1}{t}$, $t=3$ find $\frac{dw}{dt}$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\cdot\frac{dz}{dt}$$

$$= \frac{1}{z}\cdot 2\cos t(-\sin t) + \frac{1}{z}\cdot 2\sin t\cos t + \left(-\frac{x}{z^2}-\frac{y}{z^2}\cdot\frac{-1}{t^2}\right)$$

$$= \cancel{\frac{-2\sin t\cos t}{z}} + \cancel{\frac{2\sin t\cos t}{z}} \left(\frac{-x-y}{z^2}\right)\cdot-\frac{1}{t^2}$$

$$= -\frac{(\sin^2 t+\cos^2 t)}{z^2}\cdot\frac{1}{t^2} \quad -\frac{1}{1/\cancel{t^2}} \quad \frac{1}{\cancel{t^2}}$$

$$\boxed{\frac{dw}{dt} = -1}$$

Q1- $w=2ye^x-\ln z$, $x=\ln(t^2+1)$, $y=\tan^{-1}t$ $z=e^t$, $t=1$

find $\frac{dw}{dt}$ $\quad \left.\frac{dw}{dt}\right|_{t=1} = \pi+1$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

$$= 2ye^x\cdot\frac{1}{t^2+1}\cdot 2t + 2e^x\cdot\frac{1}{1+t^2} + \left(-\frac{1}{z}\cdot e^t\right)$$

$$4 \cdot \frac{(\sqrt{2})^{\cancel{2}}}{\cancel{\sqrt{2}}} - 4\sqrt{2}$$

$$= \frac{2ye^x}{1+t^2} \cdot 2t + \frac{2e^x}{1+t^2} + \left(-\frac{e^t}{2}\right)$$

$$= \frac{4tye^x + 2e^x}{1+t^2} + \left(-\frac{e^t}{2}\right) = \frac{4ye^x + 2e^x}{2} + \left(\frac{-\cancel{e^t}}{\cancel{e^t}}\right)$$

$$= \frac{4ye^x}{\cancel{2}} + e^x - 1 = 2ye^x + e^x - 1 = e^x(2y+1) - 1$$

Q/ $z = 4e^x \ln y$, $x = \ln(r\cos\theta)$, $y = r\sin\theta$, $(r,\theta) = \left(2, \frac{\pi}{4}\right)$

Find $\left.\frac{\partial z}{\partial r}\right|_{\left(2,\frac{\pi}{4}\right)} = ?$ $\sqrt{2}\,(2+\ln 2)$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial \theta} = 4\ln y\, e^x \cdot \frac{1}{r\cos\theta}\cdot r(-\sin\theta) + 4e^x \frac{1}{y} \cdot r\cos\theta$$

$$= 4e^{\ln(r\cos\theta)} \cdot \ln(r\sin\theta)\cdot\left(-\frac{\sin\theta}{\cos\theta}\right) + \frac{4e^{\ln(r\cos\theta)}}{\cancel{r}\sin\theta} \cdot (\cancel{r}\cos\theta)$$

$$= 4r\cancel{\cos\theta}\cdot\ln(r\sin\theta)\left(-\frac{\sin\theta}{\cancel{\cos\theta}}\right) + 4r\cos\theta \cdot \frac{\cos\theta}{\sin\theta}$$

$$= -4r\sin\theta \cdot \ln(r\sin\theta) + 4r\cos\theta\cdot\cot\theta$$

$$= -4(2)\sin\frac{\pi}{4}\cdot\ln\left(2\cdot\sin\frac{\pi}{4}\right) + 4(2)\cos\frac{\pi}{4}\cdot\cot\frac{\pi}{4}$$

$$= -8\frac{1}{\sqrt{2}}\cdot\ln\left(2\cdot\frac{1}{\sqrt{2}}\right) + \frac{8}{\sqrt{2}}\cdot 1$$

$$= -4\sqrt{2}\,\ln\sqrt{2} + 4\sqrt{2}$$

$$= -4\sqrt{2}\,\frac{1}{\cancel{2}}\ln 2 + 4\sqrt{2}$$

$$= 2\sqrt{2}\ln 2 + 4\sqrt{2}$$

$$= 2\sqrt{2}(-\ln 2 + 2)$$

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BSDSF24M029
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Q1 $z = 4e^x \ln y$, $x = \ln(r\cos\theta)$, $y = r\sin\theta$ $(r,\theta) = (2, \frac{\pi}{4})$

Find $\left.\frac{\partial z}{\partial r}\right|$, $\frac{\partial z}{\partial \theta}$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$= 4\ln y\, e^x \cdot \frac{1}{r\cos\theta} \cdot r(-\sin\theta) + 4e^x \cdot \frac{1}{y} \cdot r\cos\theta$$

$$= 4e^{\ln(r\cos\theta)} \cdot \ln(r\sin\theta) \cdot \left(\frac{-\sin\theta}{\cos\theta}\right) + \frac{4e^{\ln(r\cos\theta)}}{r\sin\theta} (r\cos\theta)$$

$$= 4r\cos\theta \cdot \ln(r\sin\theta)\left(\frac{-\sin\theta}{\cos\theta}\right) + 4r\cos\theta \cdot \frac{\cos\theta}{\sin\theta}$$

$$= -4r\sin\theta \cdot \ln(r\sin\theta) + 4r\cos\theta \frac{\cos\theta}{\sin\theta}$$

$$= -4(2)\sin\left(\frac{\pi}{4}\right) \cdot \ln\left(2 \cdot \sin\frac{\pi}{4}\right) + 4(2)\cot\frac{\pi}{4}$$

$$= \frac{-8}{\sqrt{2}} \cdot \ln\left(\frac{2}{\sqrt{2}}\right) + \frac{8}{\sqrt{2}} \cdot 1$$

$$= -4\sqrt{2} \cdot \ln\sqrt{2} + 4\sqrt{2}$$

$$= -4\sqrt{2}\, \frac{1}{2}\ln 2 + 4\sqrt{2}$$

$$= -2\sqrt{2}\ln 2 + 4\sqrt{2}$$

$$= \boxed{2\sqrt{2}(-\ln 2 + 2) = \left.\frac{\partial z}{\partial \theta}\right|_{(2, \frac{\pi}{4})}}$$

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$= 4\ln y\, e^x \cdot \frac{1}{r\cos\theta} \cdot \cos\theta + 4e^x \cdot \frac{1}{y} \cdot \sin\theta$$

$$\frac{\partial z}{\partial r} = \frac{4e^x \ln y}{r} + \frac{4e^x \sin\theta}{y}$$

$$= \frac{4e^{\ln r\cos\theta}\ \ln r\sin\theta}{r} + \frac{4e^{\ln r\cos\theta}\ \sin\theta}{r\sin\theta}$$

$$= \frac{4r\cos\theta \cdot \ln r\sin\theta}{r} + \frac{4r\cos\theta}{r}$$

$$= \cancel{\frac{2\cdot 8r\cos\theta \cdot \ln r\sin\theta}{r}}$$

$$= \frac{4r\cos\theta \ln r\sin\theta + 4r\cos\theta}{r}$$

$$= \frac{4r\cos\theta\left(\ln r\sin\theta + 1\right)}{r}$$

$$= \frac{4(2)\cos\frac{\pi}{4}\left(\ln 2\sin\frac{\pi}{4} + 1\right)}{2}$$

$$= \frac{\frac{8}{\sqrt{2}}\left(\ln\frac{2}{\sqrt{2}} + 1\right)}{2} = \frac{4\sqrt{2}\left(\ln\sqrt{2} + 1\right)}{2}$$

$$= \frac{\cancel{4}\sqrt{2}\left(\frac{1}{2}\ln 2 + 1\right)}{\cancel{2}}$$

$$= \cancel{2}\sqrt{2}\left(\frac{\ln 2 + 2}{\cancel{2}}\right)$$

$$= \boxed{\sqrt{2}\left(\ln 2 + 2\right)} = \frac{\partial z}{\partial r}\bigg|_{\left(2, \frac{\pi}{4}\right)}$$

Partial Derivatives with constrained variables:

Ex: Find $\frac{\partial w}{\partial x}$ at point $(x,y,z) = (2,-1,1)$ if $w = x^2+y^2+z^2$

$z^3 - xy + yz + y^3 = 1$ and x, y are indp vars. (z = dependent)

$w = x^2+y^2+z^2$

$$\frac{\partial w}{\partial x} = 2x + 0 + 2z\frac{\partial z}{\partial x} \quad \text{---(i)}$$

$$z^3 - xy + yz + y^2 + 1$$

$$3z^2 \cdot \frac{\partial z}{\partial x} - y + y\frac{\partial z}{\partial x} + 0 = 0$$

$$\frac{\partial z}{\partial x}(3z^2 + y) = y$$

$$\frac{\partial z}{\partial x} = \frac{y}{3z^2+y} \quad \text{Put in (i)}$$

$$\frac{\partial w}{\partial x} = 2x + 2z\left(\frac{y}{3z^2+y}\right) = \frac{2(2) + 2(1)(-1)}{3(1)^2+(-1)} = 4 - \frac{\cancel{2}}{\cancel{2}} = 4-1$$

$$\boxed{\left.\frac{\partial w}{\partial x}\right|_{(2,-1,1)} = 4-1 = 3}$$

$\left(\frac{\partial w}{\partial x}\right)_y$ — $\frac{\partial w}{\partial x}$ with x and y are independent

$\left(\frac{\partial w}{\partial y}\right)_{x,t}$ — $\frac{\partial w}{\partial y}$ with y, x & t are indep.

treat as constants

find $\left(\frac{\partial w}{\partial x}\right)_{y,z}$ if $w = x^2 + y - z + \sin t + \cos t$ and $x+y=t$

$$\left(\frac{\partial w}{\partial x}\right)_{y,z} = 2x + 0 - 0 + \cos t = 2x + \cos(x+y)$$

Ex: $w = x^2+y^2+z^2$ & $z = x^2+y^2$

$$\left(\frac{\partial w}{\partial y}\right)_z = 2x\frac{\partial x}{\partial y} + 2y + 0$$

$$= \cancel{2x}\left(\frac{-y}{\cancel{x}}\right) + 2y$$

$$= \cancel{-2y} + \cancel{2y}$$

$$\boxed{\left(\frac{\partial w}{\partial y}\right)_z = 0}$$

$z = x^2+y^2$

$0 = 2x\frac{\partial x}{\partial y} + 2y$

$\frac{\partial x}{\partial y} = \frac{-y}{x}$

$$\left(\frac{\partial w}{\partial z}\right)_y = 2x\frac{\partial x}{\partial z} + 0 + 2z$$

$$= \cancel{2x}\left(\frac{1}{\cancel{2x}}\right) + 2z$$

$$\boxed{\left(\frac{\partial w}{\partial z}\right)_y = 1 + 2z}$$

$z = x^2+y^2$

$1 = 2x\frac{\partial x}{\partial z} + 0$

$\frac{\partial x}{\partial z} = \frac{1}{2x}$

$$\left(\frac{\partial w}{\partial z}\right)_x = 0 + 2y\frac{\partial y}{\partial z} + 2z$$

$$= \cancel{2y}\left(\frac{1}{\cancel{2y}}\right) + 2z$$

$$\boxed{\left(\frac{\partial w}{\partial z}\right)_x = 1 + 2z}$$

$z = x^2+y^2$

$1 = 0 + 2y\frac{\partial y}{\partial z}$

$\frac{\partial y}{\partial z} = \frac{1}{2y}$

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Find $\left(\frac{\partial w}{\partial y}\right)_x$ and $\left(\frac{\partial w}{\partial y}\right)_z$ at point $(w, x, y, z) = (4, 2, 1, -1)$

if $w = x^2 y^2 + yz - z^3$ & $x^2 + y^2 + z^2 = 6$

$$x^2 + y^2 + z^2 = 6$$

$$0 + 2y + 2z\frac{\partial z}{\partial y} = 0$$

$$\frac{\partial z}{\partial y} = \frac{\cancel{2}-2y}{2z} = \frac{-y}{z}$$

$$\left(\frac{\partial w}{\partial y}\right)_x = x^2 \cdot 2y + \left(y \cdot \frac{\partial z}{\partial y} + z \cdot 1\right) - 3z^2 \cdot \frac{\partial z}{\partial y}$$

$$= 2x^2 y + z + \frac{\partial z}{\partial y}\left(y - 3z^2\right)$$

$$= 2x^2 y + z + \frac{-y}{z}\left(y - 3z^2\right)$$

$$= 2x^2 y + z - \frac{y^2}{z} + 3yz$$

$$= 2(2)^2(1) - 1 - \frac{(1)^2}{-1} + 3(1)(-1) = 8 \cancel{-1} \cancel{+1} - 3$$

$$\boxed{\left(\frac{\partial w}{\partial y}\right)_x = 5}$$

$$x^2 + y^2 + z^2 = 6$$

$$2x\frac{\partial x}{\partial y} + 2y + 0 = 0$$

$$\frac{\partial x}{\partial y} = \frac{-\cancel{2}y}{\cancel{2}x} = \frac{-y}{x}$$

~~$\left(\frac{\partial w}{\partial y}\right)_z = x^2 \cdot 2y + z - 0$~~

~~$= 2x^2 y + z$~~

~~$= 2(2)^2(1) - 1$~~

~~$= 8 - 1$~~

~~$\left(\frac{\partial w}{\partial y}\right)_z = 7$~~

$$\left(x^2 \cdot 2y + y^2 \cdot 2x \cdot \frac{\partial x}{\partial y}\right) + z = 0$$

$$2x^2 y + y^2 2\cancel{x}\left(\frac{-y}{\cancel{x}}\right) + z = 0$$

$$2x^2 y - 2y^3 + z = 0$$

$$2(2)^2(1) - 2(1)^3 + (-1) = 0$$

$$\boxed{8 - 2 - 1 = 5 = \left(\frac{\partial w}{\partial y}\right)_z}$$

Directional Derivative:

The derivative of f at the $P_0(x_0, y_0)$ in the direction of unit vector $u = u_1 i + u_2 j$ is the number

$$\left(\frac{df}{ds}\right)_{u,P_0} = \lim_{s \to 0} \frac{f(x_0 + s u_1, y_0 + s u_2) - f(x_0, y_0)}{s}$$

Provided limit exists.

The directional derivative is also denoted by $(D_u f)_{P_0}$ -

The derivative of f at P_0 in direction of u.

Ex: find the derivative of $f(x, y) = x^2 + xy$ at $P_0(\overset{x_0, y_0}{1, 2})$ in the direction of unit vector $u = \overset{u_1}{\frac{1}{\sqrt{2}}} i + \overset{u_2}{\frac{1}{\sqrt{2}}} j$

$\frac{\bar{a}}{\|a\|}$ no need to make unit vector if question says in the direction of $\bar{a}$ then find unit vector.

$$\left(\frac{df}{ds}\right)_{u,P_0} = \lim_{s \to 0} \frac{f\left(\overbrace{1 + \frac{s}{\sqrt{2}}}^{x}, \overbrace{2 + \frac{s}{\sqrt{2}}}^{y}\right) - f(1, 2)}{s}$$

$$= \lim_{s \to 0} \frac{\left[\left(1 + \frac{s}{\sqrt{2}}\right)^2 + \left(1 + \frac{s}{\sqrt{2}}\right)\left(2 + \frac{s}{\sqrt{2}}\right)\right] - (1+2)}{s} \qquad = \frac{5}{\sqrt{2}}$$

$$= \lim_{s \to 0} \frac{\left[1 + \frac{s^2}{2} + 2 + \cancel{5}3\frac{s}{\sqrt{2}} + \frac{s^2}{2}\right] - 1 + 2}{s}$$

$$= \frac{1 + \cancel{2}\frac{s^2}{\cancel{2}} + 2 + 3\frac{s}{\sqrt{2}} - 3}{s} = \frac{\cancel{3} + s^{\cancel{2}} + \frac{3\cancel{s}}{\sqrt{2}} - \cancel{3}}{\cancel{s}}$$

$$s + \frac{3}{\sqrt{2}} = 0 + \frac{3}{\sqrt{2}} = \frac{\cancel{5}3}{\sqrt{2}}$$

The Gradient vector of $f(x,y)$ at a point $P_0(x_0, y_0)$ is the vector $\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$

Theorem: The directional derivative is dot product of the partial derivative of $f(x,y)$ are defined at $P_0(x_0, y_0)$ then

$$\left(\frac{df}{ds}\right)_{u, P_0} = (\nabla f)_{P_0} \cdot \underline{u} = |\nabla f||\underline{U}| \cos\theta \qquad |\underline{U}| = 1$$

$$= |\nabla f| \cos\theta \qquad \theta: \text{angle b/w } u \text{ \& } \nabla f$$

Ex: Find the Derivative of $f(x,y) = xe^y + \cos xy$ at the point $(2,0)$ in the direction $\underline{A} = 3i - 4j$ So, vector given So, find unit vector $\sqrt{3^2+4^2}$

$$u = \frac{A}{|A|} = \frac{3i-4j}{5} = \frac{3}{5} i - \frac{4}{5} j$$

$$\nabla f = f_x i + f_y j$$

$$= (e^y - \sin xy \cdot y) i + (xe^y - \sin(xy) x)$$

$$\nabla f|_{(2,0)} = i + 2j$$

$$(D_u f)_{P_0} = \nabla f|_{(2,0)} \cdot \underline{u} = (i + 2j)\left(\frac{3}{5} i - \frac{4}{5} j\right)$$

$$= \frac{3}{5} - \frac{8}{5} = \frac{-5}{5} = -1$$

Remarks: i) The func $f(x,y)$ increases most rapidly when $\theta = 0$

$$D_u f = \nabla f \cdot \underline{U} = |\nabla f| \cos 0 = |\nabla f|$$

ii) The func $f(x,y)$ decreases most rapidly when $\theta = \pi$

$$D_u f = \nabla f \cdot \underline{U} = |\nabla f| \cos 180 = -|\nabla f|$$

iii) Any direction $\underline{U}$ orthogonal to the gradient is a direction of zero change in $f(x,y)$ bcz θ then equal $\frac{\pi}{2}$

$$D_u f = |\nabla f| \cos\frac{\pi}{2} = 0, \quad \nabla f \neq 0$$

→ For Direction of ∇f find unit vector & for how much it increases/decreases find magnitude

→ Direction of vector is its unit vector

Ex: find the direction in which $f(x,y) = \frac{x^2}{2} + \frac{y^2}{2}$

i) increase most rapidly at (1,1)

ii) decreases " at (1,1)

iii) what are the Direction zero change in f(x,y) at (1,1)

i) The func increases rapidly in direction of ∇f at (1,1)

$$\nabla f = f_x i + f_y j = xi + yj = \boxed{\underline{i} + \underline{j} = \nabla f|_{(1,1)}}$$

$$\underline{u} = \frac{i+j}{\sqrt{1+1}} = \frac{1}{\sqrt{2}}\underline{i} + \frac{1}{\sqrt{2}}\underline{j}$$

ii) The func decreases rapidly in the direction of $-\nabla f$ at (1,1)

Its direction $\underline{u} = -\frac{1}{\sqrt{2}}i - \frac{1}{\sqrt{2}}j$

iii) The direction zero change at (1,1) are directions orthogonal to ∇f

$$\underline{n} = -\frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}j \quad \& \quad \underline{n} = \frac{1}{\sqrt{2}}\underline{i} - \frac{1}{\sqrt{2}}j$$

Dot product of $\underline{u}$ & any of $\underline{n}$ is zero so $\underline{n}$ is zero change ∴

For three Variable

$$\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k} = f_x\hat{i} + f_y\hat{j} + f_z\hat{k}$$

$$(D_u f)_{P_0} = f_x|_p \underline{u_1} + f_y|_p \underline{u_2} + f_z|_p u_3$$

$$(D_u f) = f_x u_1 + f_y u_2 + f_z u_3$$

Ex: Find the Derivative of $f(x,y,z) = x^3 - xy^2 - z$ at (1,1,0) in the direction of $\underline{A} = 2i - 3j + 6k$

$$|A| = \sqrt{2^2+3^2+6^2} = \sqrt{4+9+36} = \sqrt{49}$$
$$|A| = 7$$

$$\underline{u} = \frac{2i - 3j + 6k}{7} = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$$

$$\nabla f = (3x^2 - y^2)i + (-2xy)j + (-1)k$$

$$\nabla f|_{(1,1,0)} = (3-1)i + (-2)j - k = 2i - 2j - k$$

→ In the direction of vector
→ rate of change = magnitude of vector

The Derivative of $f(x,y) = x^3 - x^2y - 2$ at $(1,1,0)$ in the direct of $\underline{A} = 2i - 3j + 6k$ is

$$\nabla f\Big|_{(1,1,0)} \cdot \underline{u} = (2i - 2j - k)\left(\frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k\right)$$

$$= \frac{4}{7} + \cancel{\frac{6}{7}} - \cancel{\frac{6}{7}}$$

$$\boxed{\nabla f\Big|_{(1,1,0)} \cdot \underline{u} = \frac{4}{7}}$$

b) In what Direction change happens most rapidly at p_0 & what are rate of change in these Directions.

most rapid change in direction of $\underline{u}$ & $-\underline{u}$

rate of change $= \nabla f = \sqrt{4+4+1} = 3$

& $-\nabla f = -3$

(★ Important)

Estimate How much the value of $f(x,y,z) = xe^y + yz$ will change if the point makes 0.1 unit from $P_0(2,0,0)$ straight toward $P_1(4,1,-2)$

we first find the derivative of f at P_0 in the direction of $\underline{u}$ the vector $\overline{P_0P_1} = (4-2)i + (1-0)j + (-2-0)k = 2i + j - 2k$

$$\underline{u} = \frac{\overline{P_0P_1}}{|P_0P_1|} = \frac{2}{3}i + \frac{j}{3} - \frac{2}{3}k$$

$$\nabla f\Big|_{P_0} = \frac{\partial f}{\partial x}\Big|_{P_0} i + \frac{\partial f}{\partial y}\Big|_{P_0} j + \frac{\partial f}{\partial z}\Big|_{P_0} k = i + 2k + 0k$$

$$\boxed{\nabla f\Big|_{P_0} \cdot u = \frac{4}{3}}$$

The change df in f that results from moving $ds = 0.1$ unit away from P_0 in the direction of $\underline{u}$ is approx

$$= \frac{4}{3} \times 0.1 = \frac{0.4}{3}$$

Tangent & Normal lines :

→ The tangent plane at point (x_0, y_0, z_0) on the level surface $f(x,y,z)=c$ is the plane through P_0 normal to $\nabla f|_{P_0}$

→ The Normal line of surface at P_0 is the line through P_0 ∥ to $\nabla f|_{P_0}$

Therefore, plane to $f(x,y,z)=c$ at $P_0(x_0,y_0,z_0)$ $(x-x_0)f_x|_{P_0} + (y-y_0)f_y|_{P_0} + (z-z_0)f_z|_{P_0} = 0$

Normal line to $f(x,y,z)=c$ at $P_0(x_0, y_0, z_0)$

$x = x_0 + f_x|_{P_0} t$

$y = y_0 + f_y|_{P_0} t$

$z = z_0 + f_x|_{P_0} t$

$(1,2,3)$ $f(x,y,z) = 4z^2 + \tan^{-1}x + y^2 = 0$

$(0,0,0)$ $f(x,y,z) = z - x\cos y - ye^x = 0$

Q| Find tangent & normal plane if $f(x,y,z) = x^2+y^2+z-9=0$ at $(1,2,4)$

Tangent plane is $(x-x_0)f_x|_{P_0} + (y-y_0)f_y|_{P_0} + (z-z_0)f_z|_{P_0}$

$(x-1)2 + (y-2)4 + (z-4) = 0$

$+2x + 4y + z = 14$

Normal Line :

$x = 1 + 2t$

$y = 2 + 4t$

$z = 4 + t$

$f_x = 2x$

$f_y = 2y$

$f_z = 1$

$f_x|_{P_0} = 2$

$f_y|_{P_0} = 4$

$f_z|_{P_0} = 1$

Theorem) Second Derivative Test for local Extreme Variables

Suppose, $f(x,y)$ and its first, second partial Derivatives are Continuous throughout a disk centered at (a,b) and that $f_x(a,b)=0$ $f_y(a,b)=0$

Then, i) $f(x,y)$ has a local maxima at (a,b) if

$f_{xx}<0$ & $f_{xx}f_{yy} - f^2_{xy} > 0$ at (a,b)

ii) $f(x,y)$ has a local minima at (a,b) if

$f_{xx}>0$ & $f_{xx}f_{yy} - f^2_{xy} > 0$ at (a,b)

iii) $f(x,y)$ has a saddle at (a,b) if $f_{xx}f_{yy} - f^2_{xy} < 0$

iv) The test is inconclusive at (a,b) if $f_{xx}f_{yy} - f^2_{xy} = 0$ at (a,b)

→ The expression $f_{xx}f_{yy} - f^2_{xy}$ is called discriminent of $f(x,y)$

EX: Find the local extreme values of the func

$$f(x,y) = xy - y^2 - x^2 - 2x - 2y + 14$$

The func is defined & differentiable for all x & y. The func therefore has extreme values only at the points where f_x & f_y are simultaneously zero

$$f_x = y - 2x - 2 = 0$$
$$f_y = x - 2y - 2 = 0$$

$$2\cancel{x} - y + \cancel{2} = 0$$
$$\pm\cancel{2x} \pm 2\cdot 2y \pm\cancel{2\cdot 2} = 0$$
$$3y + 6 = 0$$
$$\boxed{y = -2}$$

$$x - 2(-2) - 2 = 0$$
$$x + 4 = 2$$
$$\boxed{x = -2}$$

Therefor the point $(-2,-2)$ is only point where $f(x,y)$ may take on an extreme value

$$f_{xx} = -2$$
$$f_{yy} = -2$$
$$f_{xy} = 1$$

The discriminent of $f(x,y)$ at $(-2,-2)$

$$f_{xx}f_{yy} - f^2_{xy}$$
$$= (-2)(-2) - (1)^2$$
$$= 4 - 1 = 3 > 0$$

The combination $f_{xx} = -2 < 0$ & $f_{xx}f_{yy} - f^2_{xy} = 3 > 0$ tells us that $f(x,y)$ has a local maxima at $(-2,-2)$

max value of func

$$f(-2,-2) = (-2)(-2) - (-2)^2 - (-2)^2 - 2(-2) - 2(-2) + 14 = 0$$
$$f(-2,-2) = 18$$

Q Same just $f(x,y) = xy$

$f_x = y$

$f_y = x$

$x = 0$ on point $(0,0)$ func may take extreme value

$y = 0$

~~Discriminent~~

$f_{xx} = 0$ | $f_{xx}f_{yy} - f_{xy}^2$

$f_{yy} = 0$ | $0(0) - (1)^2$

$f_{xy} = 1$ | $1 > 0$